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Abstract. In human-centric technologies, skin segmentation of body parts is a prerequisite for high-level processing. The traditional method of skin detection is pixel-wise detection coupled with morphological operations. Pixel-wise methods usually generate a number of false samples and outlier skin pixels, which can make it difficult for morphological operations to provide satisfactory results in complex scenarios. Furthermore, in many cases only a coarse region is required (e.g., the bounding-box of the face) rather than detailed pixel-wise labeling. A patch-wise skin segmentation method is proposed based on deep neural networks. Our method treats image patches as processing units instead of pixels, which directly exploits the spatial information of pixels in the detection stage rather than using morphological operations on isolated pixels after detection. An image patch dataset is built and deep skin models (DSMs) are trained based on the new dataset. Trained DSMs are then integrated into a sliding window framework to segment skin regions of the human body parts. Experiments on standard benchmarks demonstrate that DSMs provide more explicit skin region of interest candidates than pixel-wise methods in complex scenarios, and achieve competitive performance on pixel-wise skin detection. © 2015 SPIE and IS&T [DOI: 10.1117/1.JEI.24.4.043009]

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1 Introduction

Skin color is a basic and powerful cue in human-centric-based computer vision applications, such as face detection¹ and vision-based human motion analysis.² Skin color has attractive properties of high-processing speed, invariance against the rotation, partial occlusion, and pose changes,^{3,4} and skin detection has been an active research area for decades.⁵⁻⁹

Skin detection is commonly used to provide region of interest (ROI) candidates for high-level processing. For example, Ban et al.¹ detected face region based on skin likelihood in YCbCr space.³⁻⁵ Guo et al.¹⁰ proposed a hand-tracking system in which skin detection was used to reduce false positive samples. Pedestrian detection¹¹ and human pose estimation¹² also benefit from skin segmentation of body parts.¹³⁻¹⁷ Dantone et al.¹⁷ combined the output of a skin detector¹⁸ with other features to train the body parts detectors. Kumar et al.¹⁵ trained a linear support vector machine (SVM) for skin color detection using red, green, and blue (RGB) values in feature extraction of parts poses. Other human parsing methods also relied on detected skin regions of body parts.^{13,14} It can be observed intuitively that only coarse skin regions are the practical requirement for these applications using skin detectors, instead of detailed pixel-wise skin classification. For example, pixel-wise skin detection of the mouth is unnecessary in face region segmentation, and rectangles of limbs will be more appropriate for human pose estimation rather than pixel-wise labeling of

limbs. This inspires us to find coarse skin regions rather than exhaustive labeling of skin pixels.

To segment skin regions from the background, the mainstream technology is pixel-wise skin detection coupled with morphological operations (i.e., erosion and dilation).¹⁹ For example, Xu et al.²⁰ used skin detectors to generate a skin map, on which dilation was used to merge subregions, and erosion was adopted to remove false candidates with small area or outlier pixels. Jin et al.²¹ adopt morphological operations to extract skin-region rectangles. However, pixel-wise skin detectors not only generate a set of discrete skin pixels, they also produce a number of false positive samples in complex background.^{3,18} The subsequent morphological operations could not present satisfactory results based on the isolated or false positive pixels. As illustrated in Fig. 1, a pixel-wise skin detector⁸ generates a set of isolated skin pixels in Fig. 1(c), in which there are many false samples (illustrated via rectangles) due to the complex background and changing illumination. Erosion and dilation [three times operations in Fig. 1(d)] could not remove outlier pixels [1st image of Fig. 1(d)], and could even aggravate the disturbance by false positive pixels [2nd image of Fig. 1(d)] and erroneously eliminate true positive pixels [3rd image of Fig. 1(d)]. In this paper, a patch-wise skin segmentation method is proposed based on deep neural networks. This method includes the following factors. (1) The method uses the image patches as the basic processing unit in classification instead of image pixels. Image patches exploit the spatial correlations of pixels, which make our method immediately

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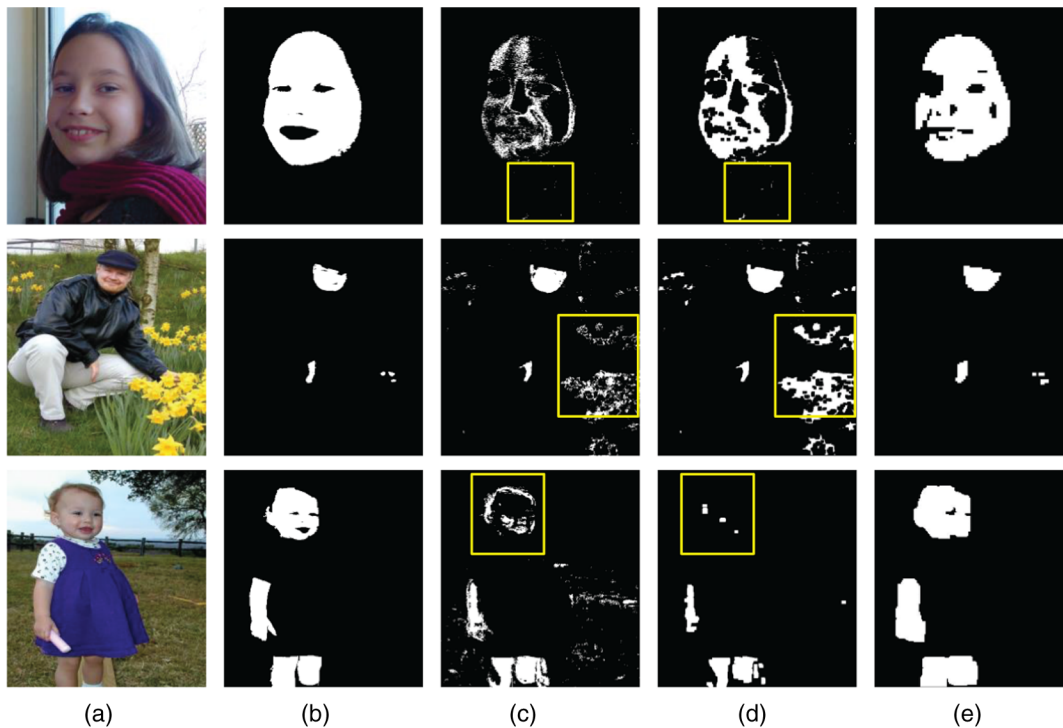


Fig. 1 Pixel-wise skin detection versus patch-wise skin segmentation. (a) The original images and (b) ground-truth images. (c) The results of pixel-wise skin detector, Gaussian mixture models (GMM).⁸ (d) and (e) The results via morphological operations (three times erosion and dilation) and the proposed patch-wise method, respectively.

segment skin regions without any postprocessing [Fig. 1(e)]. (2) A deep neural network is used in patch-wise skin detection. The structure and parameters of deep skin models (DeepSkin models, DSMs) are investigated in different color spaces. (3) A novel skin patch dataset named IRIP-patch is built for training and testing, and consists of more than 140,000 image patches with size of 20×20 , and 570,000 image patches with size of 10×10 . (4) Trained DSMs are integrated into a sliding window framework to classify image patches into skin or nonskin categories, which could obtain more explicit skin ROI candidates compared to pixel-wise methods in complex scenarios.

The remainder of this paper is organized as follows: The related work is given in Sec. 2. DeepSkin models will be presented in Sec. 3, while Sec. 4 shows the dataset and methodology. Experimental results with analysis are depicted in Sec. 5 and conclusions are presented in Sec. 6.

2 Related Work

Skin detection is conducted to build a decision rule that classifies pixels or regions as skin or nonskin. There are two main categories in skin detection: pixel-wise methods and region-based methods.^{3,4}

Pixel-wise methods classify each pixel as a skin or non-skin pixel independently from its neighbors. Many studies on pixel-based skin detection have been conducted in recent years. Jones and Rehg¹⁸ created the Compaq dataset and used a Bayesian classifier with the histogram technique (HB) for skin detection. Phung et al.³ introduced the ECU dataset and presented a study of skin segmentation with eight color spaces using HB, multilayer perceptron, Gaussian classifiers, and so on. Over the last few years, pixel-wise skin detection attracted increasing attention in computer vision. Khan et al.⁷

evaluated the performance of nine modeling techniques, such as AdaBoost, random forest (RF), and SVM. They also proposed a systematic approach for robust skin segmentation using graph cuts.²² Fusion of different color space channels was also investigated to improve the performance of pixel-wise skin detection.^{8,23} Despite its popularity, pixel-wise methods have proven to be quite challenging. For example, the choice of color space significantly affects the performance. Gaussian mixture models (GMM) in RGB color space achieved 90% true positive rate (TPR) and 15.5% false positive rate (FPR), while obtaining 90% TPR and 30% FPR in YIQ color space.^{3,4,18} The robustness of skin models is another challenge, due to the fact that many objects in the real world, such as clothes, hair, and wood in background, have skin colors; this leads the pixel-wise skin detectors to generating many false positive samples in complex background. As depicted in Ref. 3, HB and GMM achieve accuracy (ACC) of 89.79% and 85.76%, respectively, and both classifiers achieved nearly 20% FPR in RGB color space.

Skin region is a compact region of the image in which the skin pixels are spatially clustered. Region-based skin detectors attempt to take the spatial arrangement of skin pixels into account to enhance the performance. Kawulok²⁴ segmented skin region based on spatial analysis of skin pixels using the distance transform for propagating the “skinness” across the image. SanMiguel and Suja²⁵ proposed two detectors designed to get regions and isolated pixels, which were combined by morphological reconstruction filtering to keep the skin regions while removing wrongly detected regions. To increase the discriminating power of the skin classification schemes, Kawulok et al.⁹ exploited textural and spatial features for skin modeling. Considering that a pixel will belong to the skin class only if a direct neighbor does, Ruiz-del Solar

and Verschae²⁶ used neighborhood information of pixels to detect skin region through a diffusion process. Reference 27 identifies the skin-similar pixels using a generic skin-model and trained an SVM classifier using the spatial features to distinguish the correct skin Gaussian component. Classification of skin pixels is a prerequisite for most region-based methods. Efficacy of the pixel-wise classification is limited due to an overlap between the skin and non-skin pixels, which results in false positive errors, and errors would be propagated to postprocessing with spatial analysis.

Unlike the methods mentioned above, the proposed method uses image patch as the basic processing unit, which considers the spatial arrangement of skin pixels in the patch and will alleviate the disturbance problem of isolated pixels. Our method directly exploits the spatial context information of pixels in the detection stage rather than spatial analysis on isolated pixels by morphological operations after detection.

3 Deep Skin Model

Deep learning has garnered substantial attention from the machine-learning community.²⁸ Hinton et al.²⁹ showed an error rate of 1.25% in handwritten recognition by introducing the deep belief networks (DBN). Ouyang and Wang³⁰ use deep modeling to learn the visibility relationship among body parts and overlapping pedestrians in pedestrian detection. Deep learning has achieved spectacular progress in computer vision,³¹⁻³³ which inspires us to investigate its power in skin segmentation of human body parts.

3.1 Deep Neural Networks

The restricted Boltzmann machines (RBM) could be stacked and trained to form DBN.²⁹ A graphical depiction of an RBM is shown in Fig. 2. Each RBM consists of two layers, namely visible layer $\mathbf{x}$ and hidden layer $\mathbf{h}$. The connections between the nodes within each layer are restricted and $\mathbf{w}$ denotes the weights between the visible layer and the hidden layer. The joint probability distribution over $\mathbf{x}$ and $\mathbf{h}$ in RBM is

$$P(\mathbf{x}, \mathbf{h}) \propto e^{\mathbf{b}^T \mathbf{x} + \mathbf{c}^T \mathbf{h} + \mathbf{h}^T \mathbf{W} \mathbf{x}}, \quad (1)$$

where $\mathbf{b}$ and $\mathbf{c}$ are the biases of the visible layer and the hidden layer, respectively. RBMs are particular energy-based models, and interaction terms are between a hidden unit and a visible unit, but not between units of the same layer

(Fig. 2). This particular form makes it easy to compute the conditional probability distributions as

$$\begin{aligned} P(\mathbf{h}_i = 1 | \mathbf{x}) &= \text{sigm}(\mathbf{c}_i + \mathbf{w}_i \mathbf{x}) \\ P(\mathbf{x}_j = 1 | \mathbf{h}) &= \text{sigm}(\mathbf{b}_j + \mathbf{w}_j^T \mathbf{h}), \end{aligned} \quad (2)$$

where $\mathbf{w}_i$ is the row vector corresponding to the i 'th row of $\mathbf{w}$, $\mathbf{w}_j^T$ is the j 'th column of $\mathbf{w}$ and $\text{sigm}(\cdot)$ is the logistic function. More details of mathematical concepts of inference and training of RBM can be found in Ref. 28, including contrastive divergence, which is used for training RBMs to learn the parameters $\mathbf{w}$, $\mathbf{c}$, and $\mathbf{b}$. The trainable RBMs will be put together to form DBNs. An example of DBN structure is shown in Fig. 2, which consists of two stacked RBMs. The first RBM is formed between $\mathbf{x}$ (visible layer) and $\mathbf{h1}$ (hidden layer 1). Layer 2 ($\mathbf{h1}$) and layer 3 ($\mathbf{h2}$) form the second RBM. DBN models the joint distribution between the observed vector $\mathbf{x}$ and the ℓ hidden layers $\mathbf{h}^k$ as

$$P(\mathbf{x}, \mathbf{h}^1, \dots, \mathbf{h}^\ell) = \left[\prod_{k=0}^{\ell-2} P(\mathbf{h}^k | \mathbf{h}^{k+1}) \right] P(\mathbf{h}^{\ell-1}, \mathbf{h}^\ell), \quad (3)$$

where $\mathbf{x} = \mathbf{h}^0$, $P(\mathbf{h}^k | \mathbf{h}^{k+1})$ is a visible-given-hidden conditional distribution in an RBM associated with level k of the DBN. $P(\mathbf{h}^{\ell-1}, \mathbf{h}^\ell)$ is the joint distribution of the top-level RBM. Training of DBN is in a greedy layer-wise fashion. The idea is to get $\mathbf{h1}$ from the first-level RBM, then take $\mathbf{h1}$ as input for the second-level RBM to computer $\mathbf{h2}$, and so on, until the last layer. This procedure will generate parameters $\mathbf{w}_i$ and $\mathbf{c}_i$ for each layer, which can be used to initialize a multilayer neural network. These parameters then will be fine-tuned with a supervised learning criterion.

3.2 Deep Skin Model

The proposed DSM is constructed based on DBN, and Fig. 3(a) shows the graphical model of the used DBN, which consists of four layers and three RBMs. Each RBM is trained in a layer-wise greedy manner with contrastive divergence. Having pretrained RBMs, the weights and biases will be utilized to initialize the model. In skin patch classification, a label layer is added to form a feed-forward neural network with five layers [Fig. 3(b)]. Then the entire model will be fine-tuned as a feed-forward, back-propagate neural network. This procedure is equivalent to initializing the parameters of

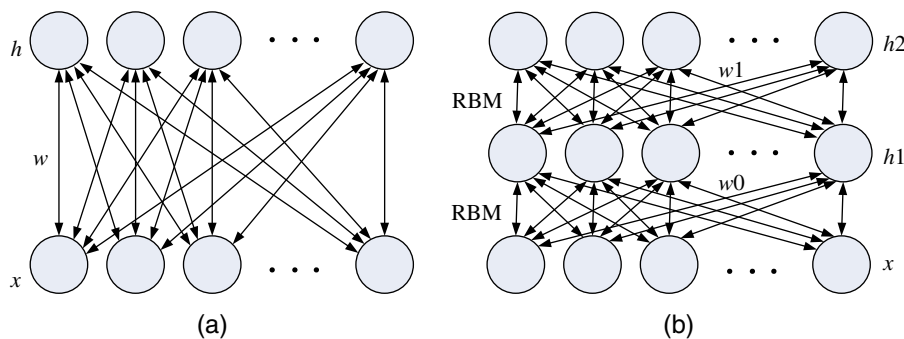


Fig. 2 (a) Graphical model of restricted Boltzmann machines and (b) a deep belief networks (DBN) model with input vector $\mathbf{x}$ and two hidden layers $\mathbf{h1}$ and $\mathbf{h2}$.

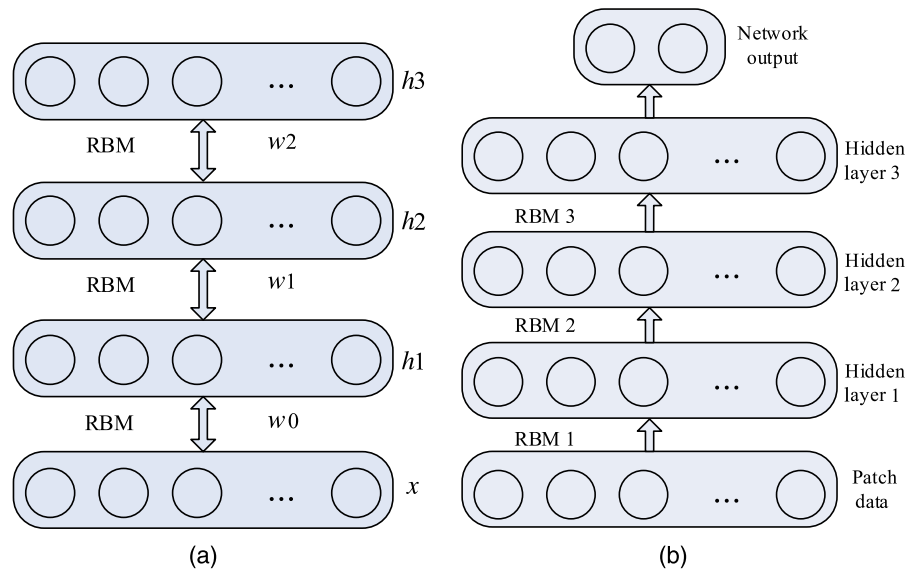


Fig. 3 (a) DBN model will be used to form the (b) deep skin model (DSM).

a deep neural network with the weights and hidden layer biases obtained by the unsupervised training strategy. There are two stages in the training of the DeepSkin model: a layer-wise pretraining and a fine-tuning stage. An image patch is expanded to an input vector $x = (x_0, x_1, \dots, x_i)$, where x_i is formed by pixel values in a specific color space (e.g., R, G, and B values in RGB color space), and i is the size of the patch. The training process of the DeepSkin model is as follows:

1. Train the first layer $h1$ as an RBM with input $x = h(0)$ as the visible layer.
2. Use $h1$ as data for the second layer.
3. Train the second layer $h2$ as an RBM, taking $h1$ as training examples.
4. Iterate (2) and (3) for the desired number of layers, each time propagating upward samples.
5. Fine-tune all the parameters with back-propagate networks in supervised learning strategy.

4 Dataset and Methodology

4.1 Dataset

The IRIP-patch dataset³⁴ is derived from ECU dataset³ and Compaq dataset.¹⁸ The Compaq dataset provides explicit nonskin images, while its ground truth data are less precise than samples in ECU. Thus, skin patches are randomly cropped from skin regions in ECU, and nonskin patches are selected from nonskin images of Compaq. Detailed information on IRIP-patch is listed in Table 1. The IRIP-patch is

Table 1 IRIP-patch dataset.

	Skin set	Nonskin set	Training set	Testing set
20× size	47,183	96,140	122,849	20,474
10× size	188,732	384,560	491,393	81,899

classified into two subsets according to the patch sizes, i.e., 20×20 and 10×10 . The image patches in each subset have been randomly divided into a training subset and testing subset. The amount of patches in the dataset is greater than 1.5 million.

Figure 4 illustrates samples in the IRIP-patch. The image patches are in a wide range of variations. For example, the first row of Fig. 4(a) shows skin patches with changing illumination and race. Skin patches in the second row show texture patterns like palm print and hair. The nose, teeth, and ears might be selected as a whole [the last two rows of Fig. 4(a)]. Nonskin patches are shown in Fig. 4(b), in which samples of the first row are very similar with skin ones while others are significantly distinct from skin ones.

4.2 Color Spaces

The choice of the color space greatly affects the performance of skin detectors. Previous work has investigated how the choice of color space and the use of chrominance channels affect skin detection.^{3,4,8,23,35} In this study, we focus on four color spaces, i.e., RGB, YCbCr, HSV, and CIELab, which are commonly used in image processing and computer vision

- RGB is very commonly used in skin detection,³⁶ since image data captured by camera are normally provided and stored as RGB;
- HSV colors are specified in hue (H), saturation (S), and intensity value (V), which can be linearly transformed from RGB;³⁷
- YCbCr separates RGB into luminance (the Y channel) and chrominance information (Cb and Cr channels). YCbCr is one of the most popular and commonly used choice for skin detection;³⁸
- CIELab is related to the RGB color space through a highly nonlinear transformation. This color space is often used for evaluation and comparison with other color spaces.³

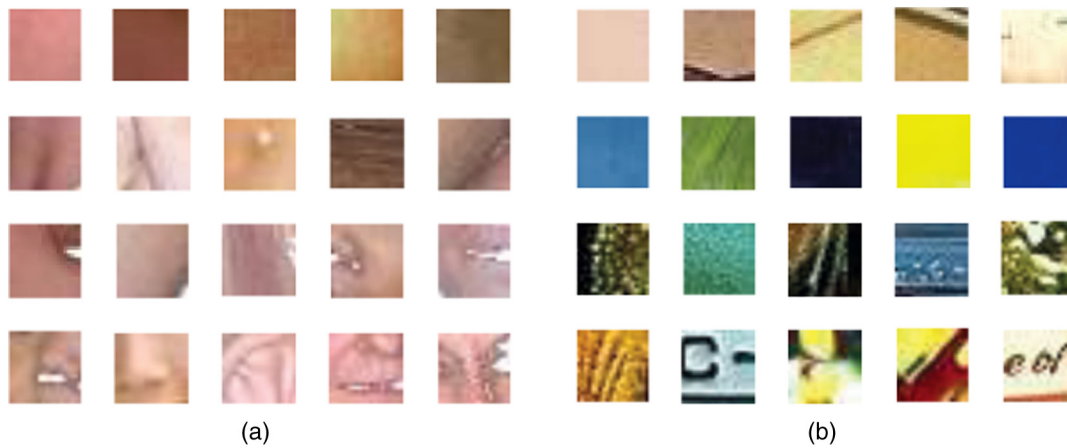


Fig. 4 Image patches in IRIP-patch dataset.³⁴ (a) The skin patches are selected from ECU dataset,³ (b) nonskin patches are selected from nonskin images of Compaq dataset.¹⁸

Although researchers argue that many existing color spaces share similar characteristics, the question of which is the optimal color space is not concluded yet. People are forced to try various color representations to make the skin model more accurate and robust. Moreover, some skin detection methods only use the chrominance by assuming that skin segmentation may become more robust to lighting variations if pixel luminance is discarded. In this paper, we will study the performance of skin patch detection on single color spaces and their combinations via DSMs.

4.3 Deep Skin Modeling

For the construction of DSMs, the DBN net is composed of three RBMs, each with 100 hidden neurons. Each RBM will be trained on the full training set with 40 epochs. Having pretrained each RBM, the weights and biases are utilized to initialize a feed-forward neural net with four layers (i.e., 100-100-100-2), in which the last two neurons are the output label units. The DSMs will be trained using mini-batches with size of 100 and 200 epochs via back-propagation. The components of RGB, YCbCr, HSV, and CIElab color spaces or their combinations in a patch are catenated to form the input vector. In the experiments, we will investigate the affections of performance on the following aspects:

- Patch size: DSM trained on “ 10×10 ” training set will be evaluated on its testing set. The model trained on “ 20×20 ” dataset will be tested on “ 20×20 ” testing set. Trained models will be integrated into a sliding window framework for skin segmentation;
- Color spaces: both the four color spaces and their combinations will be investigated in the experiments. As shown in Table 3, “RGB + HSV” means that the input vector is the catenation of three components in RGB color space and three components in HSV color space;
- Chrominance and luminance: we study whether skin patch classification may become more robust to lighting variations if luminance is discarded;

- Parameters of DBN: the performance on a different number of neurons in each layer and number of iterations will be studied in the experiments;
- Comparison with classification methods: comparisons with SVM and RF will be presented in the experiments.

4.4 Skin Segmentation via Sliding Window

Most object detectors require classifiers to evaluate image windows in a sliding window fashion.^{39,40} In this paper, trained DSMs are integrated into a sliding window framework to segment skin regions. The size of sliding window is equal to the patch size. The window will slide over the image with different sized overlaps, in which the input vector will be extracted. If the patch corresponding to the sliding window is classified as a skin patch, pixels in the window will be classified as skin ones. Roles of overlap with different sizes are further investigated in the experiments.

5 Experiments and Analysis

5.1 Experimental Setup

First, DSMs will be trained and tested on the IRIP-patch dataset.³⁴ The structure of DSMs is depicted as Sec. 4.3. Training subset and testing subset are selected randomly, which are then divided into five subsets for k -fold cross validation. Second, SVM⁴¹ and RF⁴² will be compared with trained models. In experiment, LIBSVM⁴¹ was used to train SVM classifier, in which RBF kernel and linear kernel were both investigated and grid-search was exploited to find the best parameter (i.e., C and g) using cross-validation. Investigations on color spaces and structural parameters of DBN will also be presented. The patch-wise classification performance will be evaluated in terms of TPR, FPR, and ACC. Moreover, k -fold cross-validation was used for evaluation in which k was set to 5, and the average accuracy was reported over four rounds on a PC with Intel Core i7 CPU (3.4G Hz) and 16 GB RAM. Finally, pixel-wise skin detection and an application of skin segmentation of human body parts will also be analyzed.

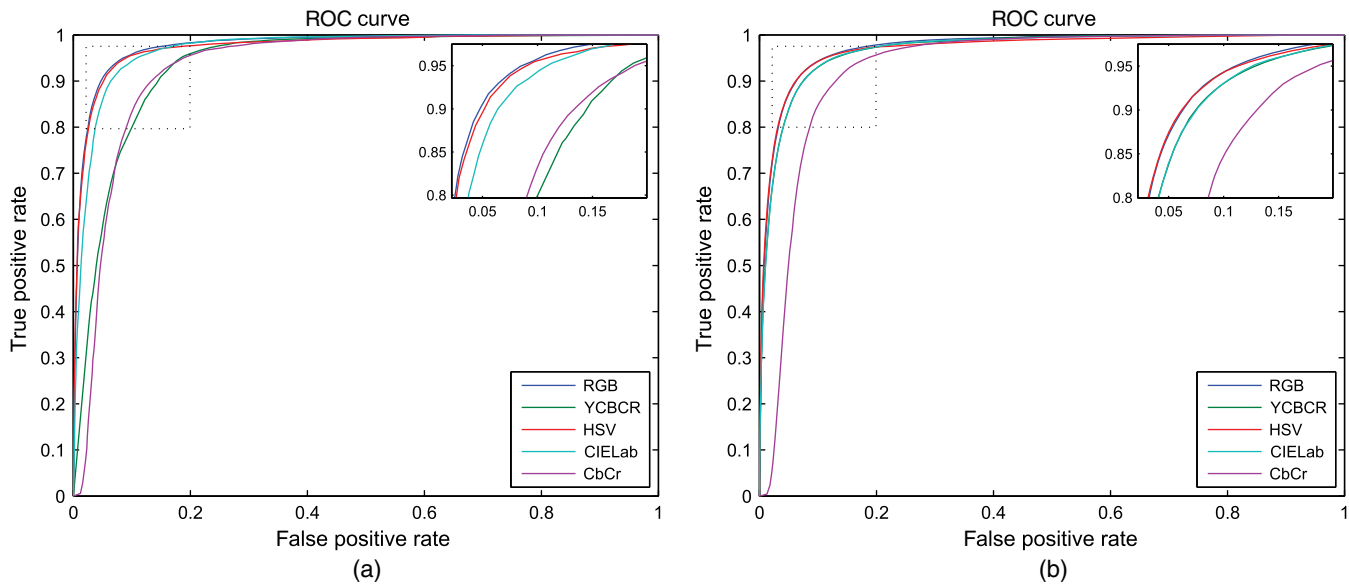


Fig. 5 The radius of curvature (ROC) curves on (a) 20×20 set and (b) 10×10 set in single color spaces. The embedded subfigures indicate the details of ROC curves.

5.2 Patch-Wise Classification of Deep Skin Models

DSMs are evaluated on the testing set of IRIP-patch. The radius of curvature (ROC) curves and the classification rates are shown in Fig. 5 and Table 2, respectively. For models trained on 20×20 datasets, the RGB model achieves 93.6% ACC, which is higher than other models. YCbCr model has 94.8% TPR and 18.3% FPR, which are both higher than others. Although the HSV model gains a lower FPR (i.e., 2.6% FPR), its TPR is also lower than others. If Y component is discarded, both TPR and FPR of models trained on 20×20 datasets decrease but not in accuracy.

Models trained on 10×10 datasets have similar FPR and ACC in RGB, YCbCr, HSV, and CIE Lab color spaces. The CbCr model has higher TPR and FPR than YCbCr model, which stands in contrast to ACC. If Y component is discarded, both the TPR and FPR of the model trained on 20×20 datasets are decreased while the ACC increased, but it is reversed for a model trained on 10×10 datasets. It cannot conclude that discarding luminance will improve the patch classification performance.

Experimental results of DSMs trained on the combinations of color spaces are shown in Table 3, and Fig. 6

Table 2 Testing results on single color space.

	20 × 20 patch			10 × 10 patch		
	TPR	FPR	ACC	TPR	FPR	ACC
RGB	91.3%	5.3%	93.6%	91.0%	6.9%	92.4%
YCbCr	94.8%	18.3%	86.1%	87.9%	6.4%	91.7%
HSV	79.2%	2.6%	91.4%	91.5%	7.1%	92.4%
CIE Lab	87.4%	5.5%	92.2%	88.7%	6.8%	91.7%
CbCr	90.4%	13.7%	87.6%	89.4%	12.6%	88.1%

presents the corresponding ROC curves. Ten combinations of four color spaces are considered. All models trained on 20×20 datasets achieve more than 91% accuracy and FPRs are restricted within a range of 2.3% and 9.9%. The models involving RGB color space (i.e., “RGB + YCbCr,” “RGB + HSV,” and “RGB + CIE Lab”) obtain better performance than others, which have similar ACC as the model of RGB. Although the model of YCbCr has a poor ACC of 86.1%, the performance of its combinations is enhanced. For example, the ACC of HSV model is 91.4%; it raises to 93.4% in. The complete combinations of all color spaces, i.e., “RGB + YCbCr + HSV + CIE Lab” (RYHC in Table 3) achieves 93% ACC, which is not the best performance. The

Table 3 Testing results on combinations of color spaces.

	20 × 20 patch			10 × 10 patch		
	TPR	FPR	ACC	TPR	FPR	ACC
RGB + YCbCr	89.5%	4.4%	93.6%	82.6%	4.0%	91.6%
RGB + HSV	88.5%	3.5%	93.6%	85.9%	4.2%	92.6%
RGB + CIE Lab	92.3%	6.0%	93.5%	83.1%	4.0%	91.7%
HSV + YCbCr	88.1%	3.9%	93.4%	83.2%	3.8%	91.9%
HSV + CIE Lab	80.0%	2.3%	91.9%	84.5%	4.2%	92.1%
YCbCr + CIE Lab	86.6%	4.9%	92.3%	82.1%	4.6%	91.0%
RGB + YCbCr + HSV	95.8%	9.9%	92.0%	78.2%	2.9%	90.9%
RGB + CIE Lab + HSV	96.2%	9.7%	92.3%	80.8%	3.3%	91.5%
YCbCr + CIE Lab + HSV	95.3%	9.9%	91.8%	81.4%	3.8%	91.3%
RYHC	93.3%	7.0%	93.1%	90.0%	6.2%	92.5%

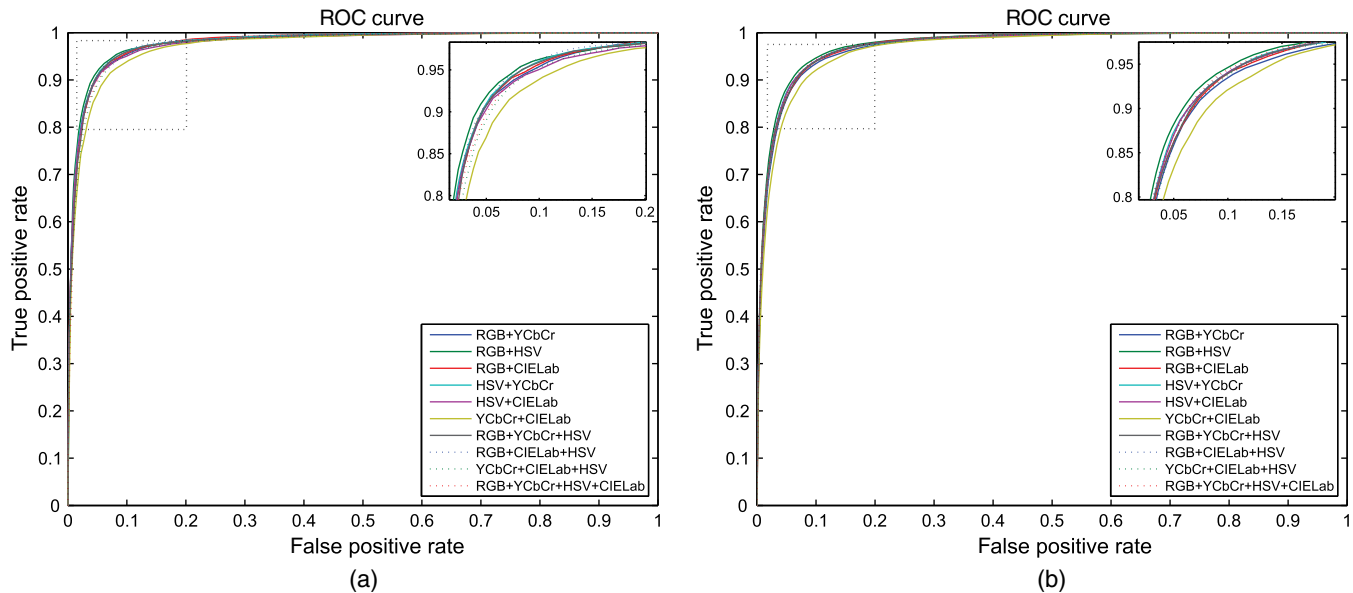


Fig. 6 The ROC curves on the combinations of color spaces. (a) The ROC curves on 20×20 patch set and (b) the ROC curves on 10×10 patch set. The embedded subfigures show the details of ROC curves.

performance of models on 10×10 combinations are slightly decreased compared to models of single color space. More color spaces in combinations does not mean improved performance for DSMs; this may be attributed in part to the property of self-feature learning for deep neural networks.³⁰

5.3 On Number of Neurons and Iterations

Initializing the parameters manually is a key step in the training stage of deep neural networks. In experiments, the performance on different numbers of neurons in hidden layers is investigated with RGB color space on 20×20 datasets. With 100, 200, and 500 neurons, the DSM achieves an ACC of 93.56%, 93.3%, and 93.59%, respectively.

Figure 7(a) shows the ROC curves of each iteration on different numbers of neurons. More neurons in the hidden layer cannot significantly improve the performance. The performance on different numbers of iterations was also investigated with RGB color space on 20×20 datasets. Figure 7(b) shows the ROC curves. Performance of more iterations will be a bit better while more time-consuming in the training stage.

5.4 Comparisons with Support Vector Machines and Random Forests

For evaluation of performance on skin patch classification, DSMs are compared with SVM⁴¹ and RF⁴² on RGB and “RGB + YCbCr + HSV” (RYH) color spaces on 20×20

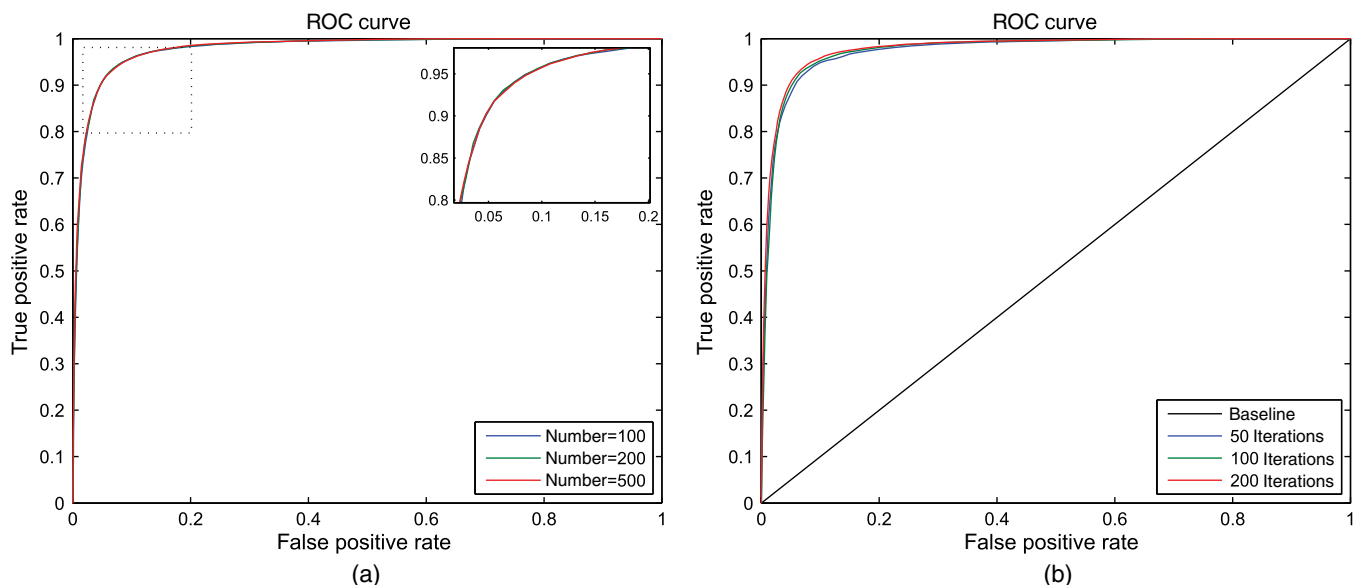


Fig. 7 The ROC curves on (a) number of neurons and (b) number of iterations.

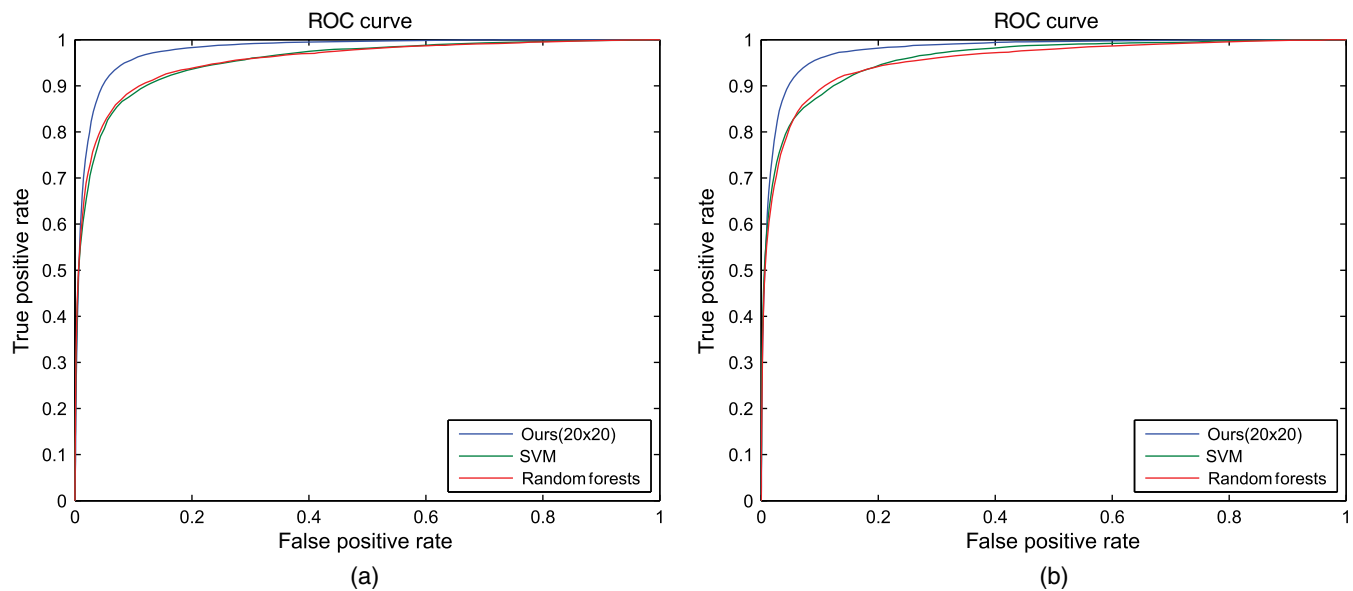


Fig. 8 The ROC curves of comparisons with support vector machines, random forest, and ours on (a) RGB color space and (b) the combinations of RGB, YCbCr, and HSV. The training and testing are based on the 20×20 dataset.

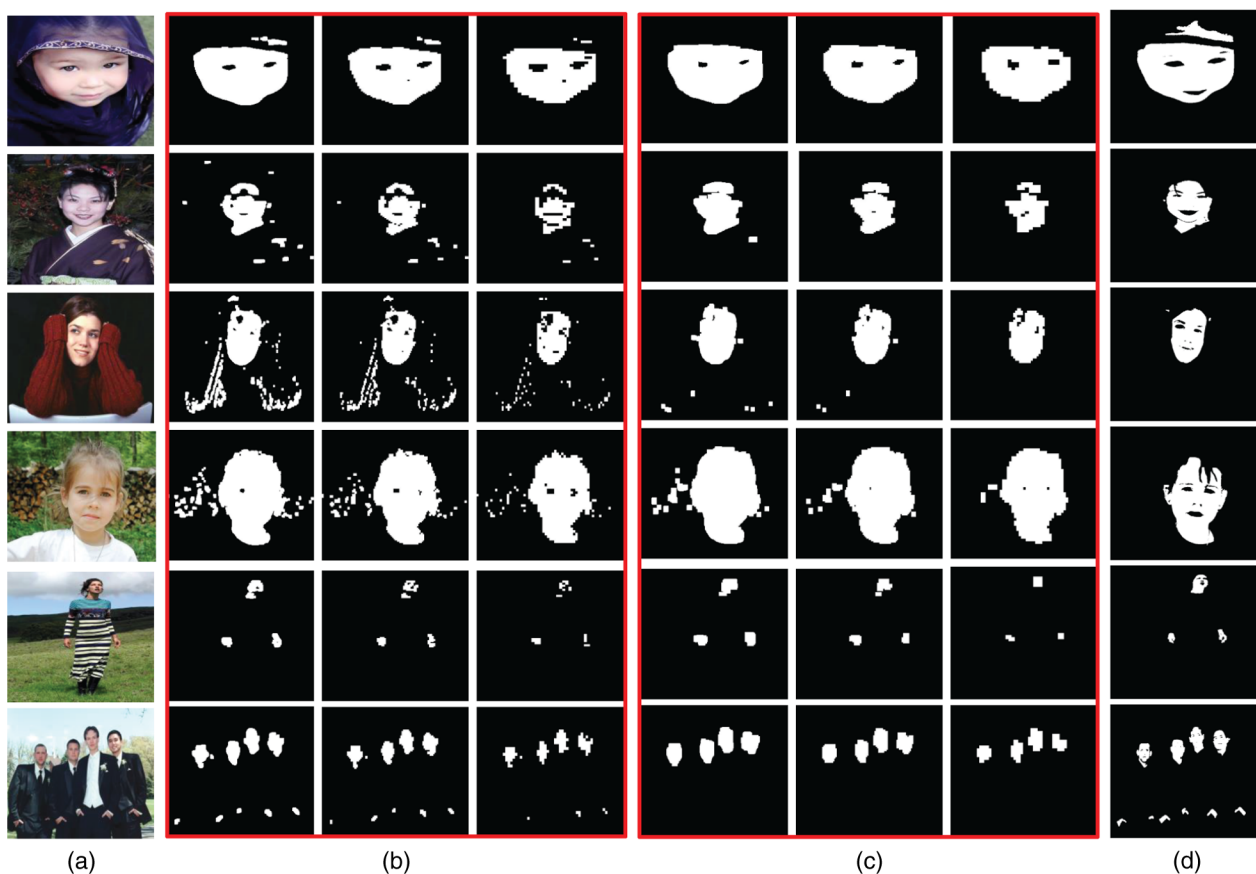


Fig. 9 Skin segmentation results via different DSMs. (a) The original images and (d) the ground-truth. (b) Segmentation results of DSM₁₀², DSM₁₀⁵, and DSM₁₀¹⁰, from left to right and (c) the results of DSM₂₀², DSM₂₀⁵, and DSM₂₀¹⁰, respectively.

Table 4 Comparisons with pixel-wise skin detectors.

	HB	GMM	FNT	DSM ₁₀ ²	DSM ₁₀ ⁵	DSM ₁₀ ¹⁰	DSM ₂₀ ²	DSM ₂₀ ⁵	DSM ₂₀ ¹⁰
TPR	83.3%	80.8%	87.1%	93.8%	91.6%	85.1%	95.1%	93.9%	90.4%
FPR	11.3%	15.8%	9.6%	12.5%	12.1%	10.1%	14.6%	11.8%	10.6%
TNR	71.4%	68.1%	73.0%	69.4%	71.7%	73.7%	69.1%	71.4%	72.4%

datasets. Linear kernel and RBF kernel were both investigated. RBF kernel was selected to train the SVM classifier because it can handle the case when the relation between class labels and attributes is nonlinear and achieve better cross-validation rate. Grid-search on C and g using cross-validation is adopted, finding the optimal parameters C and g in the range of $[2^{-5}, 2^5]$ in experiment. The step size of finding C and g is set to 0.5 and the remaining parameters are exploited by default values of LIBSVM. The RF uses linear weak learners with 100 trees. Figure 8 shows the ROC curves of the testing results. SVM obtained 89.4% ACC on RGB and 90.5% ACC on RYH. The ACC of RF on RGB and RYH are 87.8% and 89.3%, respectively. DSM shows better performance in patch-wise classification.

5.5 Skin Segmentation via Sliding Window

For segmentation of skin ROI, the trained DSMs are integrated into a sliding window framework. A subwindow with size equal to the patch size slides from top to bottom and left to right in an image. In experiments, DSMs trained on 10×10 datasets and 20×20 datasets in RGB color space are utilized to detect skin regions with step size of 2, 5, and 10, respectively. The DSMs will be denoted by DSM_m^n for convenience, where m represents $m \times m$ dataset (e.g., 10×10 dataset) and n indicates step size of the sliding window. Thus, DSM_{10}^2 means the model is trained on a 10×10 dataset in RGB color space and will be used in sliding window framework with step of 2 pixels. Skin segmentation results of different DSM are illustrated in

Fig. 9. Comparisons with pixel-wise skin detectors were also investigated.

5.5.1 Skin segmentation results of different deep skin models

Skin segmentation results of different DSM models are illustrated in Fig. 9. As shown in the first row, all DSM models achieve satisfactory face region detection, though fail to capture details of the mouth compared to ground truth. In complex scenes with illumination changing (2nd row and 3rd row) and objects having skin-tone color (4th row), DSM_{20} model presents better face segmentation than DSM_{10} model, but produces poor details of eyes and mouth. Even in wild and remote scenes, as shown in the last two rows, DSM models demonstrate robust and high performance in detection skin ROI of human body parts. Although DSM_{20} model cannot capture skin regions with small area (e.g., hands in the last row), it provides more explicit face regions. The step size of the sliding window plays an important role in skin detection. For example, images in the fourth row show a progressive filtering process of cluttered background with step sizes of 2, 5, and 10 pixels, respectively. DSM model with a larger step size is more robust to clutter background, but results in more false samples due to jagged edges.

5.5.2 Comparison with pixel-wise skin detection methods

Performance of pixel-wise skin segmentation of DSMs is further investigated. In the experiments, DSM models and

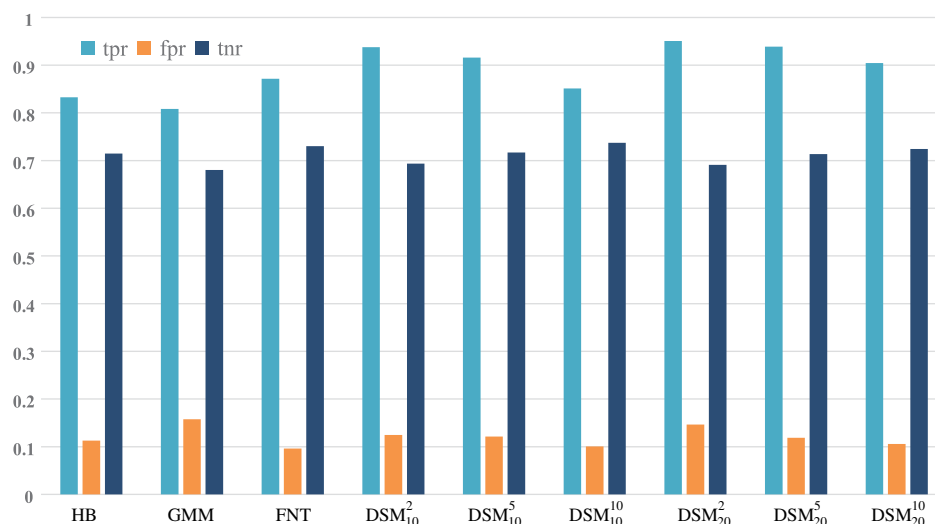


Fig. 10 Skin pixel classification of pixel-wise skin detectors and DSM models.

pixel-wise skin detectors are both evaluated on an ECU dataset. Table 4 shows the results in terms of pixel-wise skin detection. A histogram-based Bayes classifier (HB)¹⁸ is trained on ECU with histogram size of 256 bins per channel in RGB color space; GMM uses parameters published by Ref. 3; flexible neural tree model (FNT) is referred to Ref. 8. Figure 10 shows the diagram in terms of TPR, FPR, and the rate of true negative (TNR). DSM models achieve higher TPR than pixel-wise methods excepting the DSM₁₀ model, which obtains the highest TNR of 73.7%. Because DSM models focus on coarse skin region detection rather than exhaustive pixel-wise classification, the FPR of DSM models are higher than FNT, but slighter better than HB and GMM. DeepSkin models have high TNR, which demonstrates that DSM models have strong power of segmenting

skin region from a cluttered background. For detailed skin detection, DSMs could not present explicit details of eyes, mouth, and hands, which results in higher FPR.

DSM models achieve more explicit skin regions compared with pixel-wise skin methods. Figure 11 shows skin segmentation results of pixel-wise skin detectors and DSM models. As shown in the first row, pixel-wise methods present rough face regions via isolated pixels, while DSM method directly obtains the face regions. Morphological operations may be helpful for pixel-wise methods to achieve skin ROI candidates, however it will fail in scenarios as shown in the second row. Both pixel-wise methods and DSMs suffer from a complex background and changing illuminance, as shown in the third row and fourth row, while DSM method could reduce the false samples

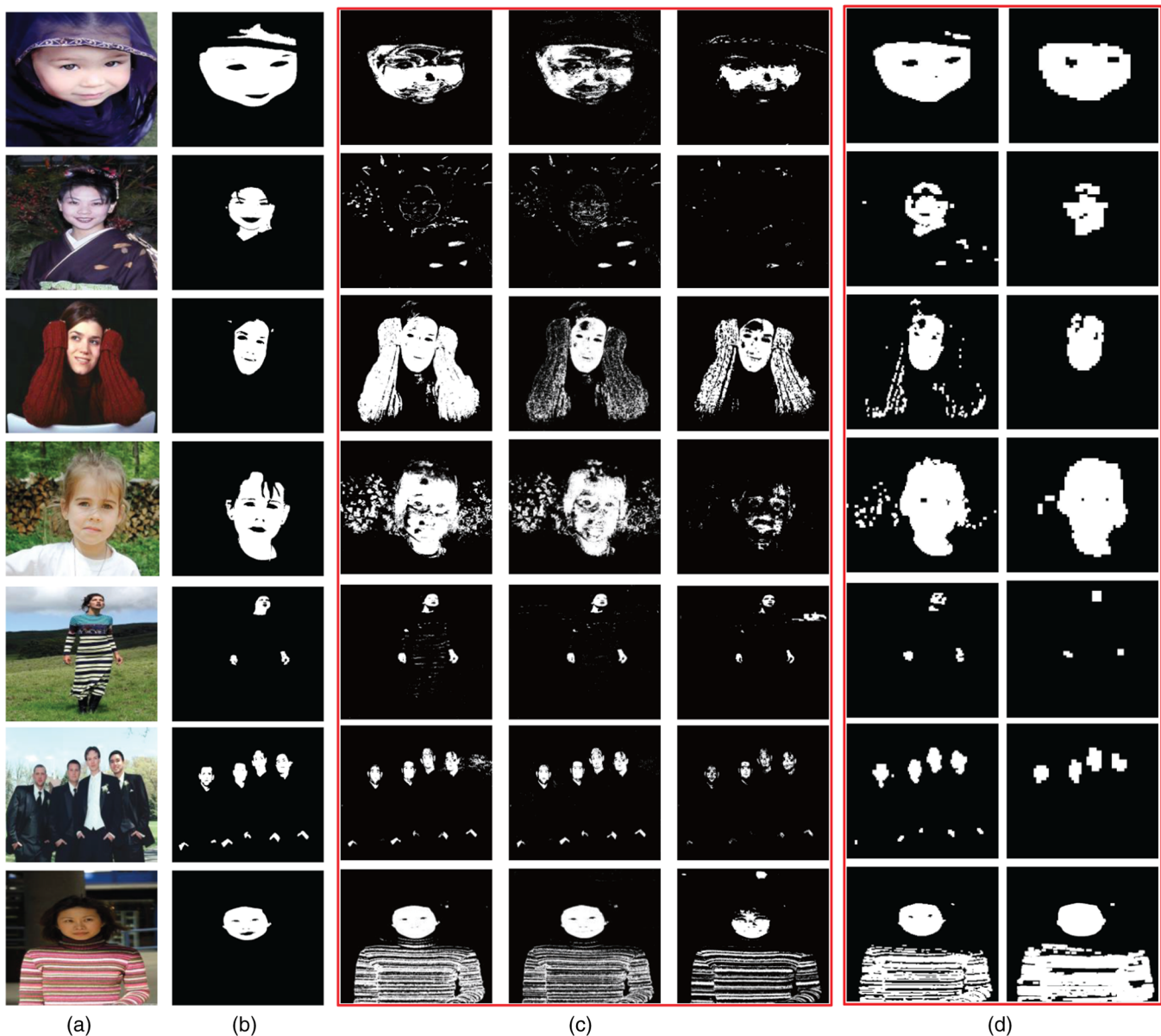


Fig. 11 Skin segmentation results of pixel-wise skin detectors and DSMs. (a) The original images and (b) the ground-truth, (c) segmentation results of GMM,³ histogram-based Bayes classifier (HB),¹⁸ and flexible neural tree (FNT)⁸ from left to right, and (d) the results of DSM₁₀⁵ and DSM₂₀¹⁰, respectively.



Fig. 12 More results of skin segmentation. (a) Original images and ground-truth and (b) the results of GMM, HB, and FNT from left to right, (c) skin segmentation results of DSMs, DSM_{10}^2 , DSM_{10}^5 , DSM_{10}^{10} , DSM_{20}^{10} , DSM_{20}^5 , and DSM_{20}^{10} .



Fig. 13 Skin segmentation of human body parts on LSP dataset.⁴⁴

significantly and present the primary face regions. For detection of human body parts with low resolutions, such as original image in the fifth row, the skin regions are very small. The false positive pixels will be negative for the identification of skin ROI. The challenges with pixel-wise methods also exist in skin detection of DSMs. For example, all methods treat the frock as skin candidates in the last row. Considering the detailed skin detection, DSMs are not as effective as pixel-wise methods. For instance, DSM method obtains the face region with no mouth in the first three rows, while pixel-wise methods could find the eyes and mouth. More experimental results can be found in Fig. 12.

5.5.3 Skin segmentation of human body parts on LSP dataset

Body part detection is critical in pictorial structure-based human pose estimation.^{16,43} Skin detector could present skin ROI candidates of body parts for pose inference. For example, skin detector provides coarse regions of face, arms and legs, as well as hands and joints of the human body. In our work, the proposed DSM is applied to skin segmentation of human body parts on an LSP dataset.⁴⁴ As illustrated in Fig. 13, our method provides explicit skin regions without postprocessing. The locations of the skin region of body parts can be obtained directly via computing the minimal bounding rectangle.

6 Conclusions

The ultimate goal of using skin detectors is to present explicit skin ROI candidates for further processing. In this paper, a patch-wise skin segmentation method is therefore proposed

to meet this goal. Image patch, not pixels, is the elemental processing unit in our method. The DSMs are trained and tested on a novel IRIP-patch dataset. Our method directly provides explicit skin ROI candidates without any morphological operations in complex scenarios. For performance of patch classification, the DSM achieves better performance than SVM and RF on IRIP-patch. Our method is patch-wise skin detection method and shows robust performance for skin segmentation in cluttered scenes. Patch size and step size of the sliding window play key roles in DeepSkin, which will significantly influence the skin segmentation results. Deep learning has shown excellent power with large-scale datasets; more training samples or convolutional neural networks with different sizes of image patches will be investigated in future work.

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